NITHISH KUMAR R

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LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

AI & Data Science graduate (2026) focused on Python-based LLM applications, RAG, AI agents, and cloud deployment. Built local-first AI assistants, retrieval-backed learning systems, and cloud-hosted ML applications. Experienced with FastAPI, SQL databases, Docker, and Google Cloud.

TECHNICAL SKILLS

AI / ML: Generative AI, LLM Applications, RAG, Agentic Workflows, Prompt Engineering, Semantic Search, Vector Databases, LangChain, Scikit-learn, PyTorch, CatBoost, XGBoost, LightGBM
LLM / AI Tooling: Gemini API, OpenAI API, Google ADK, Ollama, Qdrant, pgvector, FAISS
Backend & APIs: FastAPI, Flask, REST API, Pydantic, SQLAlchemy, OAuth 2.0 / PKCE, JWT Authentication
Frontend: Next.js, Electron.js, Streamlit
Databases: PostgreSQL, pgvector, MySQL, SQLite, Qdrant
DevOps & Cloud: Docker, Google Cloud Platform (Cloud Run, Cloud Storage, OAuth), Git, GitHub, Vercel, Virtual Machines

PROJECTS

Robin - A Personal Assistant for everyone - Live Demo *Present*
Python, FastAPI, Electron.js, React, TypeScript, SQLite, Qdrant, Ollama, Google ADK, WebSockets
- Built a Windows desktop AI assistant that handles conversations, reminders and daily tasks using local Ollama or cloud Gemini models, with persistent memory and sessions.
- Integrated Gmail, Google Calendar, Tasks, files and browser tools using OAuth and approval-gated AI actions, so sensitive changes execute only after user review.
- Added real-time voice interaction, wake-word detection and streaming responses; verified 932 backend and 52 desktop tests and packaged a Windows installer.

LearnTrack — AI Learning Continuity Platform - Live Demo *June-2026*
Python, FastAPI, PostgreSQL, pgvector, Gemini API, Next.js, Docker, Google Cloud Run
- Built an AI learning companion that captures study notes or voice, lets users review AI-extracted topics and questions, and remembers next steps across sessions.
- Added PostgreSQL/pgvector search, evidence-linked RAG answers and personalized study checkpoints through FastAPI services, JWT authentication and a Next.js UI.

Food Delivery ETA Prediction - End-to-End ML Pipeline - Live Demo *May-2026*
Python, CatBoost, Scikit-learn, XGBoost, LightGBM, Streamlit, Docker, Google Cloud Run
- Built an ML pipeline on 41,953 delivery records using order, traffic and weather features; compared five regression models, with CatBoost reaching $R^2 = 0.84$.
- Created a Streamlit app that predicts delivery time from user inputs using reusable preprocessing and inference; packaged with Docker and deployed on Google Cloud Run.

RESEARCH & PUBLICATIONS

AI-Driven and NLP-Powered Personalized Education Tutor for Various Languages of the Indian Region - Vaishnavi J., Prasanth K. R., **Nithish Kumar R**., Pragadheesh M., Haresh K.
8th International Conference on Intelligent Sustainable Systems (ICISS 2026), IEEE-affiliated | ***Paper ID: ICISS-0103*** | *Accepted*

EDUCATION

M. Kumarasamy College of Engineering, Karur, Tamil Nadu *2022 - 2026*
B.Tech in Artificial Intelligence and Data Science | CGPA: 7.33 / 10

Velammal Matric. Hr. Sec. School, Theni - Tamil Nadu State Board *2022*
Higher Secondary Certificate | 70%

Directorate General of Military Training, Bangalore - CBSE *2020*
Secondary School Leaving Certificate (Military) | 54%

CERTIFICATIONS

- **IBM AI Developer Professional Certificate** - Generative AI, Prompt Engineering, Python, Flask (2024)
- **NPTEL** - The Joy of Computing Using Python (Score: 72%)
- **NPTEL** - Ethical Hacking